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Alzheimer's Disease Detection using Ensemble Learning and Artificial Neural Networks

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Abstract. This paper presents an ensemble method using machine learning classification algorithms and an artificial neural network-based scheme using the popular and widely used open access series of imaging studies (OASIS) dataset for Alzheimer's disease (AD) detection. The proposed work performs an in-depth feature examination and a training-test split in a $70:30$ ratio on the dataset and applies 8 different ML algorithms. The AD detection outcome is obtained using two procedures, first by an ensemble approach applied to different machine learning algorithms, and secondly by using an artificial neural network (ANN). The use of ANN achieves an overall test accuracy of 0.9196 whereas two ensemble techniques, namely gradient boosting and voting classifier achieve an overall test accuracy of 0.857 and 0.8304. The precision and sensitivity scores demonstrate the superior detection performance of the ANN over the ensemble method on ML algorithms.

Keywords: Alzheimer's disease detection · Machine learning · ANN · Ensemble learning

1 Introduction

1.1 Context

Alzheimer's disease (AD) is a type of dementia that affects a person's memory, thought, and behaviour. It is a disease that begins with mildly affecting parts of the brain, which makes an individual have difficulty in remembering any newly learned information, constant changes in mood, and confusion with events, times, and places. AD usually starts in an individual after the age of 60 years and its risk increases as the person ages. So the early detection of the disease can help the doctors start the treatment which can significantly help alleviate some symptoms, reduce their intensity, and thus contribute to a better quality of life for patients and their families. In recent years, various MRI biomarkers

have been proposed for grouping AD patients in various illness stages. Despite numerous endeavours, distinguishing effective AD-explicit biomarkers for the early determination and expectation of ailment movement is still awaited and requires more research.

1.2 Related works

Ensemble learning and neural networks have increased various individual results that are obtained from non-linear network designs [1,2]. Various conventional AI strategies, for instance, SVM and feedforward neural networks are used effectively to analyze AD through MRI pictures as given in [3,4]. One such ongoing technique is [4], where a dual-tree complex wavelet transform is utilized to extricate features, and a feed-forward neural network is utilized to characterize the pictures. Expound conversation and 1167 similar outcomes with other famous conventional strategies can likewise be found. A mix of patches extricated from an autoencoder followed by convolutional layers for feature extraction was utilized in [5]. While these profound learning strategies give great precision results, our proposed method for determining the training information and utilization of transfer learning, we get equivalent and in fact slightly better accuracy while utilizing just a small amount of the training information and computation. The information mining strategies in an MRI signal give an approach to separate the useful information from the MRI data comprising a huge number of voxels. The insignificant arrangement of voxel values is important to accomplish an adequately high precision for the expectation and conclusion of Alzheimer's Disease. Multivariate methodologies, for example, PCA [6], ICA [7], SEM [8], and SVM [9] are potential applicants yet have not been applied to several useful neuroimaging information up until this point. As of late, such multivariate strategies have been received for the examination of basic MRI to distinguish spatial examples of decay in Alzheimer's Disease [10–12]. The use of such classifiers of a spatial example of decay in MCI has indicated promising outcomes for the expectation of Alzheimer's Disease [10] [11]. Ding et al. [13] validated an algorithm to cater diagnosis of Alzheimer's disease with PET scans of brain images with a ROC curve of 0.98.

1.3 Problem identification

Recently, different imaging tools for Alzheimer's disease detection have been proposed by impactful research findings [14–17]. Some works [18] propose segmentation and identification frameworks to differentiate Alzheimer's disease from mild cognitive impairment. Some have worked towards stratification of Alzheimer's by creating a risk-score neural network model [19].

Currently, different imaging and screening procedures using deep learning tools for feature extraction like convolutional neural networks (CNNs), deep CNNs, or recurrent neural networks (RNNs) are being widely used. Each learning algorithms extract patterns using convolutional, pooling, activation functions/layers, batch normalization layers, and multiple other layers, depending

upon the type of network. The aforementioned procedure processes around millions of trainable points (trainable parameters). Note that the size of the training data is a primary factor for the total number of trainable parameters.

Therefore, the above-mentioned process differs from machine learning approaches in terms of computational efficiency. Traditional machine learning models (SVM, kNN, Gradient boosting, etc) efficiently classify patterns using statistical approaches with lowered computation time.

1.4 Contribution outline

This paper proposes a comparative analysis between machine learning algorithms and a multi-layered perceptron or artificial neural network (ANN) when trained on a classification dataset. The problem statement is currently a topic of utmost significance due to the challenges faced by subjects exposed to Alzheimer's disease or other dementia-related disorders.

Towards this, the proposed AD detection scheme consists of two different approaches. Firstly, the model is trained with conventional machine learning architectures, and then two ensemble learning architectures termed Gradient Boosting and Voting Classifier are used to get the decision of the class to which it belongs. The gradient Boosting algorithm consists of results acquired from the decision tree classifier and random forest classifier whereas the voting classifier considers the results from the SVM, XGBoost, Adaboost, random forest, and decision tree classifiers. Furthermore, an artificial neural network (ANN) architecture is also trained and used for AD detection, and the results are evaluated using evaluation parameters. The workflow is demonstrated in Fig. 1. The following section gives a highlight of the experimentation conducted with the OASIS dataset (373 subjects).

2 Experiments

2.1 Data and pre-processing

For our experiments, the training and testing examples comprising MRI reports and other information about AD patients have been collected from the popular open access series of imaging studies (OASIS) dataset [20]. It contains two sets of data containing cross-sectional MRI information in young, moderately aged, Non-demented, and demented established grown-ups, and the longitudinal MRI information of the Non-demented and grown-ups. The first set has a cross-sectional assortment of 416 subjects aged 18 to 96 years. The second set comprises a longitudinal assortment of 150 subjects aged 60 to 96 years. Classes in the independent categories were ID (patient ID), M/F (male or female), Hand (which hand is more dominant, L/R), Age, Educ (level of education), SES (socio-economic status), MMSE (mini-mental state examination), CDR (clinical dementia rate), eTIV (estimated total intracranial volume) and nWBV (Normalize whole brain volume) (see 2.2 for more about MMSE, CDR,

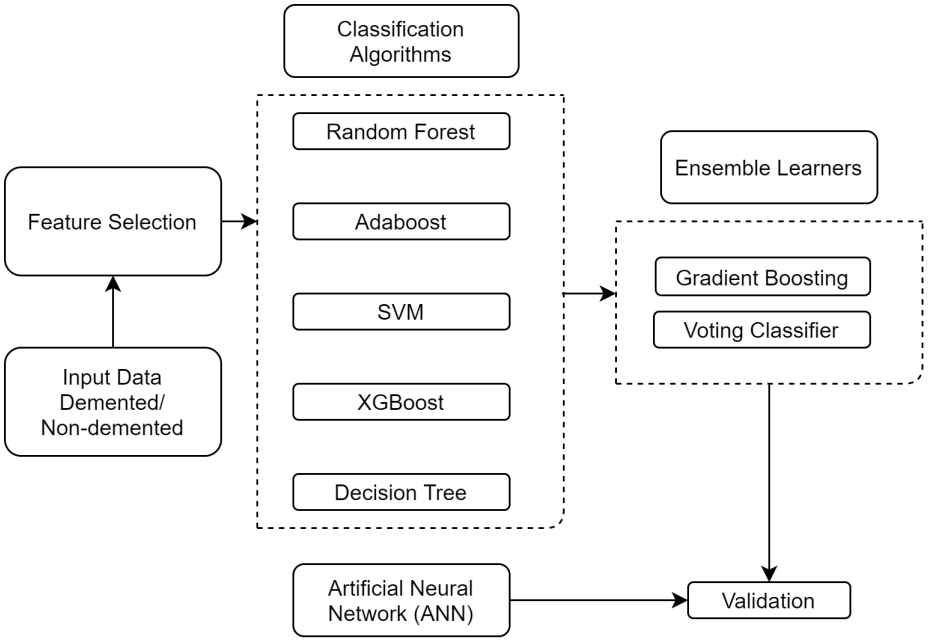


Fig. 1: Proposed workflow

eTIV, nWBV, and ASF) along with target column (Non-demented / Demented) were assessed from the OASIS dataset. The data contains 373 examples and 15 features. Violin plots of CDR by Gender and Age are shown in Fig 2a and 2b, respectively. Label Encoder was used to encode the columns into numeric data values, test size was set to 30% (112), keeping 70% (261) for training the classifiers.

2.2 Abbreviations

The classes taken for experimentation are discussed in brief:

(a) The Mini-Mental State Examination (MMSE) or Folstein test is a 30-point survey that is utilized broadly in clinical and exploratory settings to quantify subjective disability.
(b) The Clinical Dementia Rating (CDR) is a 5-point scale used to portray six spaces of intellectual and practical execution appropriate to Alzheimer's infection and related dementias: Memory, Orientation, Judgment and Problem Solving, Community Affairs, Home and Hobbies, and Personal Care. The vital data to make each evaluation is acquired through a semi-organized meeting of the patient and a solid witness or guarantee source (e.g. relative). The CDR table gives observations that control the clinician in making proper validations, dependent on information and clinical evaluations.

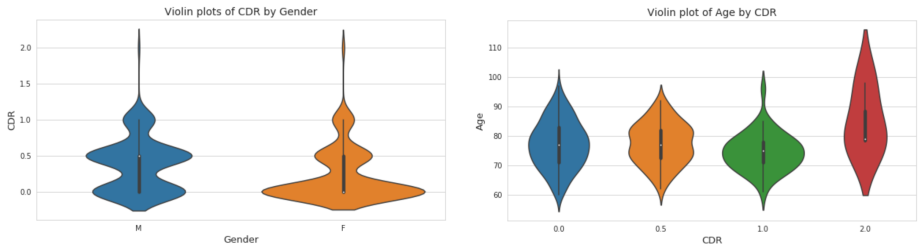
(c) Estimated Total Intracranial Volume (eTIV) is efficiently used as a proxy to maximize brain volume (pre-morbid), sometimes accounting for cognitive reserve [21].
(d) Normalize whole brain volume (nWBV) shares an estimate of the relative percentage of the intracranial cavity that is occupied by the brain.
(e) Atlas Scaling Factor (ASF) is a methodology for morphometric and useful information investigation in young, old, and demented adults utilizing mechanized atlas-based head size standardization.

2.3 Model architectures

The proposed work compares two learning frameworks. One is the ensemble learning applied over several ML algorithms and the other is an ANN-based classification scheme.

Machine Learning algorithms The popular Random Forest Classifier, SVM, Decision Tree Classifier, XGBoost, Voting Classifier, Gradient Boosting Classifier, and Adaboost Classifier were used for the disease classification. In Random Forest Classifier, the max depth was set to 8, keeping the number of estimators as 200. For SVM, C (regularization parameter) was set to 0.1 by applying hyperparameter tuning. In the Decision Tree Classifier, the max depth was set to 5. For the XGBoost classifier, regularization parameters were appropriately set for min child weight, gamma, subsample, colsample by tree, and max depth. Voting Classifier was applied on all these algorithms with the individual weights and parameters, as were functional on the individual classifiers. Feature importance for all the classifiers was evaluated along with ROC curves, accuracy, precision, and sensitivity. Another ML framework, Adaboost combines multiple stumps who are technically weak learners thus increasing their learning strength. The number of stumps was set to 60 signifying decision trees have one node and two leaves. Adaboost assigns equal sample weights to all records which are then fed into a model and based on the performance of each stump, weights of incorrectly classified records are increased and correctly classified are decreased. Gradient boosting is an ensemble boosting technique that basically optimizes the classification loss function. Decision trees used in gradient boosting are comparatively larger in size than Adaboost. Voting Classifier uses the technique of ensemble learning using which the results of various ML models operating simultaneously are merged together. This helps in reducing over-fitting and improving accuracy. Therefore, it combined the outputs of machine learning algorithms like Random Forest, Adaboost, XGBoost, Decision Tree and SVM. The final class prediction was done using hard voting.

Artificial neural networks ANNs are computational networks comprised of highly interconnected neurons that work together simultaneously to analyse and process information similar to the way the human brain does. It consists of dropout layers, dense layers and activation functions stacked together. We used



(a) (b)

Fig. 2: Violin plots of CDR with (a) Age and (b) Gender

a dense layer consisting of 8 neurons. A dropout rate of 0.5 was used which randomly disables half of the neurons of a layer which helps in reducing over-fitting and the training time required for each epoch. Activation functions are differentiable functions that help to decide which of the neurons will get activated and will push forward their values onto the next layer. They provide neurons with the ability to learn complex and complicated data. Adam optimizer [22] was used to update weights and bias. The Sequential ANN architecture is considered as follows: Dense$(8) \Rightarrow$ Dropout$(0.5) \Rightarrow$ Activation['ReLU'] $\Rightarrow$ Flatten $\Rightarrow$ Dense$(1) \Rightarrow$ Activation['Sigmoid'].

The individual machine learning models were trained with the described specifications and the ANN model was trained with 500 epochs. The validation accuracy converged at 0.92, suggesting the highest performance.

2.4 Evaluation protocol and performance metrics

The equations for accuracy, precision and sensitivity are given as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{2}$$

$$\text{Sensitivity} = \frac{TP}{TP + FN} \tag{3}$$

2.5 Results analysis

The models were first evaluated with respect to the individual architectures, and ensembled architectures are then compared to the ANN architecture. The models used are compared in Table. 1 with respect to their Accuracy, Precision, and Sensitivity. ANN outperforms other models, with an accuracy of 0.9196, with a misclassification value of 0.0804. This signifies that the model evaluates

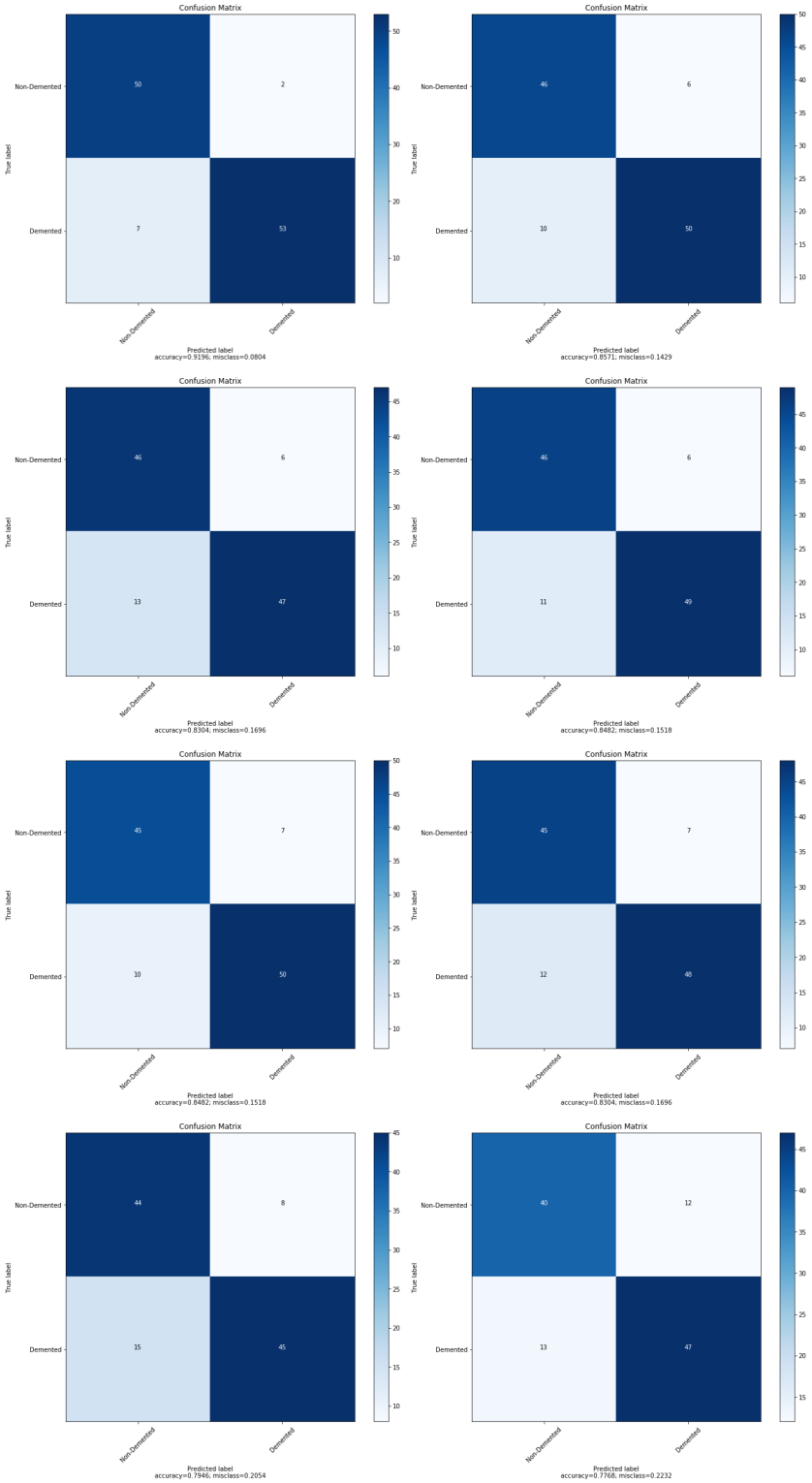


Fig. 3: Confusion matrices for (a) ANN, (b) Gradient boosting, (c) Voting classifier, (d) Random forest, (e) Adaboost, (f) Decision tree, (g) XGBoost, and (h) SVM.

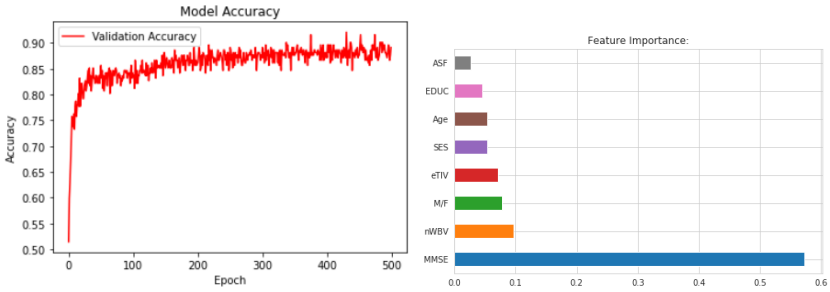


Fig. 4: (a) ANN training: 500 epochs (b) Feature importance analysis.

Table 1: Performance Evaluation of all architectures used.

Classifier	Accuracy	Precision	Sensitivity	F-1 Score
ANN	0.92	0.962	0.877	0.917
Gradient Boosting	0.857	0.85	0.857	0.851
Voting Classifier	0.830	0.885	0.78	0.828
Random Forest	0.848	0.85	0.85	0.844
Adaboost	0.848	0.80	0.80	0.841
Decision Tree	0.830	0.86	0.86	0.825
XGBoost	0.795	0.846	0.746	0.792
SVM	0.777	0.769	0.755	0.761

103 test cases correctly, out of 112. The ensembled architectures, namely Gradient Boosting and Voting Classifier show accuracy close to 0.857 and 0.8304, respectively. The confusion matrix in Fig. 3 shows the true positives and false negative rates, which further determine the Accuracy, Precision, and Sensitivity of all the algorithms used in our experiment. The accuracy, precision, and sensitivity scores for the classifiers are mentioned in Table. 1 and are calculated from the equations as mentioned.

Fig. 4 (a) demonstrated validation accuracy of the model after 500 epochs. The validation set was kept the same as the test set. Feature importance maps shown in Fig. 4 (b), demonstrate that the MMSE parameter (around 0.6) highly accounts for the target column, i.e, Demented/Non-demented. This ensures that slight changes in MMSE can re-evaluate the class of the subject, where other columns are given lesser importance (around 0.1).

The ANN classifier outperforms the ensemble and individual machine-learning classifiers due to the following reasons:

1. The performance of the multi-layer perceptron architecture (ANN) can be optimized by a variable number of epochs, governing validation of model convergence, when training and validation losses are monitored.
2. The regularization of the cost function of machine learning algorithms is concluded after a certain number of attempts of tuning the concerned hyper-parameters.

3. The performance of the ANN architecture is expected to optimize when the multi-labelled features are data engineered with more correlation.

3 Conclusion

The proposed scheme was experimented with and analysed using the ensembled learners on ML algorithms and the ANN model. The ANN model outperforms the rest of the defined classifiers. Furthermore, investigation and analysis of different methods for predicting Demented and Non-demented labels are aimed as a future scope, along with the increase of data, to precisely address the Alzheimer's Disease problem.

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